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The design of adaptive communication frame supporting high-speed transmission based on ModBus protocol

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Abstract

In the era of industrial big data, the demand for high-speed data transmission is higher and higher. The traditional Modbus protocol uses fixed data frame length transmission, which can not make full use of link resources, resulting in very low transmission rate. In the early stage, in order to solve the problem of traditional ModBus transmission protocol, an algorithm was proposed to adjust the frame length by calculating the optimal frame length through the frame error rate. However, because the frame length of each data frame is adjusted in this algorithm, the link stability becomes worse and the efficiency is reduced. In order to solve this problem, this paper proposes an improved adaptive frame length adjustment algorithm, which adjusts the frame length according to the retransmission failure rate per unit time, and gives different adjustment schemes according to different situations, which improves the link stability. Finally, this paper takes the servo drive data acquisition system as the carrier, and verifies the effectiveness of the algorithm through experiments.

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Keywords: Serial communication; ModBus transmission protocol; adaptive frame length; servo driver

1. Introduction

With the advent of the Industry 4.0 era, emerging industries such as big data and cloud computing have entered the industrial field. Big industrial data driven by digitalization has promoted the development of manufacturing industry to a new mode. The subject of communication is no longer limited to people or people and things. More

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and more material-oriented communication services are used in industrial scenes, such as automatic control in factories, industrial internet of things, automatic driving, etc., which poses a huge challenge to the efficiency of data transmission in industry [1]. Servo drive system, as an important part of industrial automation, has the advantages of high precision and fast response in the field of motion control. However, it still cannot meet the demand of data transmission in the application scenarios where a large amount of data needs to be transmitted quickly.

According to the above problem, This paper will optimize and improve the ModBus protocol, and design a serial communication protocol suitable for high-speed data transmission, which can support the transmission of adaptive data frames. After the upper computer sets a baud rate, the communication protocol can dynamically adjust the length of transmitted data frames according to the retransmission failure rate of incorrect data per unit time. Compared with the traditional ModBus communication protocol, this adaptive communication frame can effectively avoid data congestion, which can not only improve the transmission rate but also improve the success rate of data transmission.

Through the introduction of ModBus communication frame format, this paper analyzes the shortcomings of the traditional ModBus protocol in the big data transmission scenario, and in the third part, it optimizes these shortcomings and gives an improved adaptive communication frame algorithm. Finally, the servo drive is used as the carrier to verify the above theory and algorithm through experiments.

2. Introduction to ModBus communication frames

ModBus protocol is a communication protocol that supports single master and multiple slaves. Master devices can communicate with slave devices alone, or with all devices in the form of broadcast. Modbus can be used as an application layer protocol on TCP/IP networks or for serial link connection. There are two transmission modes in the latter, namely ASCII (American Standard Information Exchange Code) mode and RTU (Remote Terminal Unit) mode [2]. ASCII mode is that each 8-bit byte will be sent as 2 ASCII characters. The main advantage of this mode is that characters can be sent at intervals of up to 1 second without errors. RTU, on the other hand, transmits data directly in the original hexadecimal format, which can transmit more data at the same baud rate than ASCII. Therefore, it is more appropriate to adopt RTU mode in application scenarios requiring high-speed data transmission [4].

In RTU communication frame, Modbus message is placed in the message frame with known starting and ending points, and the starting and ending points of the message must be clearly defined. Since RS232 is sent in a single byte in serial communication, the number of bytes in the data field is highly uncertain, so the integrity of the message must be detected and the wrong identification can be set [5,6]. Use RTU mode when sending a message, at least to begin with a pause interval of 3.5 characters of time, because the entire message frames must be as a continuous flow to lose. If there is a pause of more than 1.5 characters before the frame completes, the receiving device refreshes the incomplete message and assumes that the next byte is the address field of a new message. Similarly, if a new message starts after the previous message in less than 3.5 characters, the receiving device will consider it a continuation of the previous message. A typical RTU message frame is shown in Figure 1.

beginning			
≥3.5 character			
address	function	data	CRC check
8bits	8bits	N*8bits	16bits
ending			
≥3.5 character			

Fig. 1. RTU message frame format

The data domain contains the data required by the terminal to perform a specific function or the data collected by the terminal in response to a query. The contents of the data may be values, reference addresses, or limits. In the case of reading input states, the query information specifies the starting address of the input to be read and the number of input signals.

Address	Fun	D1 start address hi	D1 start address lo	D1 number hi	D1 number lo	CRC16 hi	CRC16 lo
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Fig. 2. RTU data query format

The state of each input port in the response information is shown in Figure 3, corresponding to each bit value in the data area respectively. The first data word is the addressing address in the query, and the other input ports are arranged from low to high in the byte in order until they reach 8 bits. The eight input bits in the next byte are also sorted from low to high. If the number of input digits returned is not a multiple of 8, the remaining bits in the last data byte are filled with zeros up to the highest bit of the byte. The byte count area describes the number of bytes of total data.

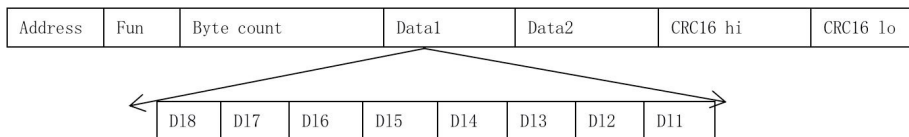


Fig. 3. RTU data response format

3. Design of serial communication frames

3.1. Design scheme of traditional adaptive frame

As the transmission of serial communication is in an environment with interference and instability, there will be a certain frame error rate. Traditional adaptive communication frame algorithms change the length of the communication frame by calculating the frame error rate, thus changing the transmission rate dynamically. For example in the field of short-wave communication. In literature [7], DRC algorithm and Trinder DRC algorithm in the original STANAG 5066 standard [5] are improved, and the false frame rate judgment is increased before changing the data rate, so as to reduce the delay of data transmission. In the field of serial communication, a method to change transmission efficiency based on frame error rate was proposed long ago [8,9]. However, the traditional adaptive algorithm based on the calculation of frame error rate has two problems. Firstly, the calculation of the transmitted data frame error rate will lead to the vibration of transmission rate and decrease of link connectivity. Secondly, to calculate the transmission rate, it is necessary to know the correct data frame in advance, which is difficult to do in the scene of industrial data acquisition. The ModBus protocol USES CRC(cyclic redundancy check code)[10,11] to solve the second problem. The CRC value is calculated by the transmitting device and then attached to the data frame. The receiving device recalculates the CRC value when receiving data, and determines whether an error has occurred by comparing it with the value in the received CRC field. In order to solve the first problem, this paper proposes an overall solution, which can adjust the data frame length adaptively according to the retransmission rate per unit time.

3.2. Design of self-adjusting communication frame algorithm

From the above Modbus communication frame, we can see that the length N contained in the data field is variable. When considering the transmission rate and reliability, we need to use a frame with a specific length.

Generally, in error channel, the longer the data frame is, the more error prone it is, which will lead to long retransmission data and increase the time consumption; If the data frame is too short, both sides will be in the state of waiting and replying frequently, which not only reduces the channel utilization rate, but also increases the communication time, making the transmission time longer^[8]. Therefore, when choosing an appropriate frame length, we should consider the influence of frame length on the frame error rate and the relationship between frame length and channel utilization^[12,13]. It is proposed that the channel throughput is T^* , the frame length is L , the channel bit error rate is BER , and the channel baud rate is R . The relationship between them can be described by the following function:

$$T^* = f(L, BER, R, \infty) \quad (1)$$

Where is some other parameters, such as retransmission times, processing time of data frames, etc. The optimal frame length can be obtained by solving the differential equation by mathematical formula:

$$\frac{\partial f}{\partial L} = 0 \rightarrow L_{opt} \quad (2)$$

In order to avoid too frequent oscillation of transmission rate, the algorithm is required to ensure the stability of transmission over a period of time and determine the frame length quickly in a short time.

The improved frame length adaptive algorithm is designed as follows:

- Firstly, the frame length L of the initial transmitted data frame is determined according to the set baud rate. ModBus protocol stipulates that the maximum length of the data frame is not more than 256 bytes. Except some control information, the length of the data is not more than 252 bytes.
- The receiving end determines whether the data is correct through CRC check. If the data is not correct, the receiver will send a retransmission request. If the data retransmission of more than 50% within one second is still incorrect, the length of the data frame will be reduced to 1/2 of the original, but not less than the set threshold.
- If more than 50% of the data is successfully transmitted in one second, the length of the data frame becomes twice as long as the original one, but it can not exceed the maximum value
- If the data frame length is between 1/2 and 2 times of the original data frame length for more than 2 times, the data frame length becomes L_{opt} .

In this way, we can not only avoid and reduce the frame length of data in the case of blocked links, but also make full use of link resources and speed up data transmission in the case of good link conditions. At the same time, the problem of repetitive oscillation in a range of data in real-time adaptive algorithm is avoided.

4. System experiment and result analysis

In this paper, the traditional ModBus protocol and the improved ModBus protocol are analyzed with the servo driven data acquisition system^[14,15] as the carrier. In this experiment, the request data frame is sent through the system, and the servo motor sends the data frame containing the CRC check field after receiving. After receiving, the system changes the CRC check value through a pseudo-random experiment to simulate the error of the communication check result.

Experiment 1 compares the transmission rates of the ModBus protocol and the improved ModBus protocol with the baud rate of 9600 for data frames between 1s and 7s under the condition of smooth link and solid link respectively. Set the amount of data in the initial data frame to 15, and the experimental results are shown in Figure 4. It can be seen that in the case of smooth links, the transmission rate of the improved ModBus protocol will gradually become faster, while that of the traditional ModBus protocol tends to be stable. However, in the case of link congestion, the improved ModBus protocol has more obvious advantages. When the link congestion occurs, the

transmission rate of the traditional ModBus protocol will always decline to a stable transmission rate, while the improved ModBus protocol will adjust according to the congestion situation.

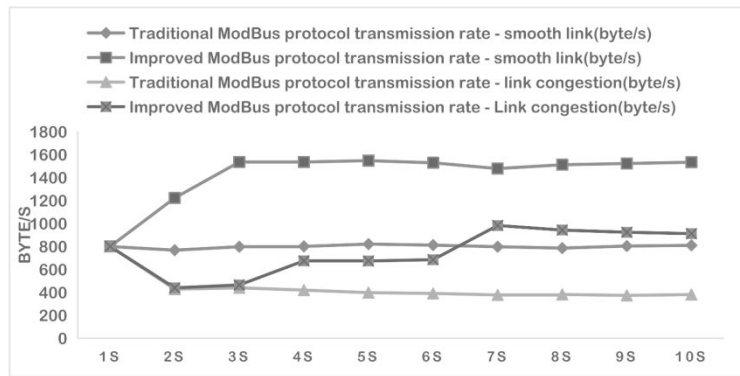


Fig. 4. Comparison of transmission rates between the traditional ModBus protocol and the improved ModBus protocol

In experiment 2, a comparison is made between the traditional adaptive algorithm and the improved adaptive algorithm on the link stability in the case of congestion. Baud rate 9600 is still taken as an example and data frames are taken as units. As shown in Figure 5, within 1s of link communication, the improved ModBus protocol tends to be stable, but the traditional adaptive ModBus protocol has obvious oscillation, and the transmission stability becomes worse, which affects the transmission efficiency.

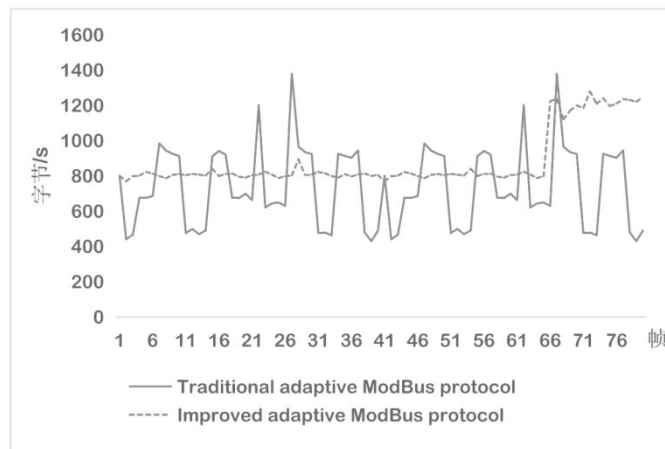


Fig. 5. Comparison of stability between the traditional adaptive algorithm and the improved adaptive algorithm

5. Conclusion

In this paper, an improved frame length adaptive algorithm is proposed, and two comparative experiments are carried out with the servo drive data acquisition system as the carrier and the Baud rate of 9600 as an example. It is verified that the improved adaptive ModBus protocol can make better use of link resources than the traditional transmission protocol, and the transmission efficiency is higher in the case of congestion. In addition, the improved adaptive ModBus protocol has a more stable link and more efficient transmission than the traditional real-time adaptive algorithm based on frame error rate. Therefore, the algorithm proposed in this paper is more suitable for

high-speed data transmission in the era of big data, which can not only ensure the stability of transmission, but also make full use of link resources.

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